

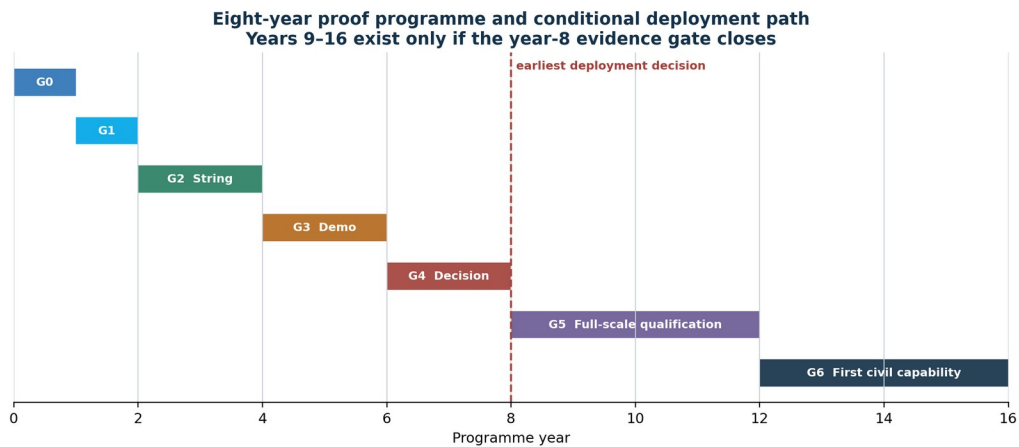
# ARGUS-PD/POD-7K Programme Capstone

*Operational Plausibility, Build Sequence, Cost Exposure, Governed Control and Strategic Space Advantage*

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Standalone strategic programme study • Independent preprint

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proof by year 8 • direct mode \$18.1B • P80 \$27.1B • governed strategic advantage

## Abstract

The first three ARGUS-PD/POD-7K papers establish a controlled theoretical programme: a complete moving-mass boundary and projectile family; a fixed seven-kilometre integrated architecture; and a coupled feasibility model with uncertainty propagation and falsification criteria [1–3]. They do not establish that a full system can presently be built, afforded, deployed or authorised. This programme capstone addresses that gap without converting a research concept into an operational counterspace plan. It asks what a government–industry consortium would have to build, in what order, at what indicative cost, under which control regime, and with what credible strategic outcome.

The analysis reaches four principal findings. First, eight years is a defensible horizon for a decision-grade research programme, not for assuming an operational seven-kilometre spacecraft. By year eight, an unusually well-funded programme could plausibly deliver validated component models, representative pulse and moving-interface experiments, an integrated ground-based demonstrator at a deliberately reduced scale, and a non-energetic orbital assembly-and-control pathfinder. A first supervised civil capability is more credibly placed in a conditional 12–16-year window. The full reference architecture remains unschedulable until mass, pulse-power, thermal, structural, projectile-survival, in-space assembly and legal authorisation gates close.

Second, commercial batteries can be useful buffer assets but do not solve the pulse-power problem. The 514.286 GJ reference nameplate equals approximately 36.5 current Tesla 4-hour Megapacks by stored energy, but their combined continuous output is only about 35.7 MW against a 30.6 TW mean admitted-power screen during the 14 ms reference launch: a power-rate gap of about  $8.57 \times 10^5$ . Sandia's 22 MJ Z facility is an important pulsed-power analogue, yet the ARGUS nameplate is about  $2.34 \times 10^4$  times larger by stored energy [7–9]. This is not evidence of impossibility; it is evidence that batteries, intermediate stores, switches, transmission, cells, structure and controls must be qualified as distinct tiers.

Third, a line-by-line re-audit reconciles my own earlier estimate with programme risk. The mutually exclusive direct work packages have a \$18.1 billion most-likely sum. After sampling their three-point ranges and a separate integration-and-reserve factor, the constant June-2026-dollar screen produces a P50 of \$24.3 billion and a P80 of \$27.1 billion for the eight-year proof programme. The risk-loaded mathematical input envelope is \$10.1–48.45 billion. These are research-planning assumptions, not quotations or an acquisition baseline; NASA and GAO practice requires a complete work-breakdown structure, supplier data, schedule logic and joint cost-and-schedule confidence before commitment [10–13]. Full deployment is deliberately not given a false point estimate at this maturity.

Fourth, the strategic hypothesis requires precise but direct language. The competition is already public: the United States seeks space superiority; China describes a journey towards a strong space presence; the European Union seeks resilience, strategic autonomy and the role of a global space power; the United

Kingdom seeks to preserve strategic advantage; NATO treats space as an operational domain; and India is expanding sovereign and commercial space capacity [24,28,33–37]. United States Space Force doctrine defines space superiority as freedom to operate at a chosen time and place without prohibitive interference while denying the same to an adversary [24]. ARGUS alone cannot establish that complete condition. If its technical gates close, however, the programme could confer substantial strategic space advantage through pulsed-power competence, launch and in-space-assembly capacity, space-domain awareness, materials science, logistics, standards, alliance leadership and control of a high-value planetary-defence service. Those benefits are strongest when the capability is transparent, civil-led, independently supervised and internationally consulted. An overt national “supremacy machine” would increase counterspace ambiguity, vulnerability, balancing behaviour and legal risk, potentially destroying the governance advantage it sought to create.

The investable proposition is therefore a stage-gated national research programme with commercially valuable spillovers and explicit stop rules. Funding buys verified knowledge and enabling infrastructure at every gate; continued scale-up is earned by evidence.

**Keywords:** programme architecture; planetary defence; pulsed power; in-space assembly; technology readiness; cost uncertainty; space governance; strategic advantage; dual-use systems.

## 1. Executive proposition

**Decision proposition.** *Fund an 18-month formulation tranche that creates the independent authority, complete work-breakdown structure, digital thread, test plan and supplier evidence needed for Gate G0. Do not authorise full-scale procurement, orbital energetic operation or a claim of deployability. Preserve the right to narrow, pause or terminate every non-closing branch.*

This is a proposal for controlled capability creation rather than a promise that the current reference system will work. Its value is asymmetric. If the central mechanism eventually closes, the sponsor acquires a strategically important civil-space infrastructure capability. If it does not, the sponsor still obtains reusable assets in pulsed power, dynamic materials, hypervelocity diagnostics, autonomous assembly, precision metrology, high-assurance control and planetary-defence modelling.

The initial authorisation should have a ceiling of approximately \$1.0–1.8 billion over 18 months. It should procure evidence, not a production design. Release of later tranches should require:

1. experimentally measured performance against pre-registered thresholds;
2. an independent technical and safety review;
3. a refreshed cost-and-schedule confidence analysis;
4. legal authority and continuing-supervision arrangements; and

5. public or internationally shareable evidence sufficient to distinguish a civil research programme from covert counterspace preparation.

The programme's most credible strategic message is simple: a sponsor intends to lead the industrial and governance architecture for planetary defence and very-large space infrastructure. It is not asking others to trust an unverifiable claim of benign intent.

## 2. What the existing research establishes

### 2.1 Controlled inheritance from Papers I–III

The capstone treats the prior papers as controlled inputs rather than marketing collateral.

| Prior work         | What it establishes                                                                                                                                | What remains unestablished                                                |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Paper I [1]        | complete 0.300 kg moving-mass boundary; projectile family; energy–momentum Pareto trade; material and flight-environment variables                 | survivable moving article; selected flight material; release accuracy     |
| Paper II [2]       | fixed 7,000 m architecture; ideal force-area-field bounds; 6,000 physical module-bay hypothesis; energy, thermal, structure and governance ledgers | representative cell; physical module closure; structure and cost baseline |
| Paper III [3]      | coupled mass-area-energy iteration; covariance; sensitivity; evidence levels; quantitative stop rules                                              | integrated hardware; validated target response; mission authorisation     |
| Launcher study [4] | variable-energy modes; physical module interpretation; rear counter-impulse arithmetic; dust, atmosphere and application envelopes                 | buildable launcher design; trajectory survival; operational mission plan  |

For the reference moving assembly,

$$E_k = \frac{1}{2} m v^2 = 150 \text{ GJ}, p = m v = 300 \text{ kNs}, \quad (1)$$

and, for constant acceleration over  $L = 7,000 \text{ m}$ ,

$$t_L = \frac{2L}{v} = 14 \text{ ms}, \dot{F} = \frac{E_k}{L} = 21.429 \text{ MN}. \quad (2)$$

At an assumed electrical efficiency of 35% and a 20% installed reserve, the reference admitted energy is 428.571 GJ and nameplate energy is 514.286 GJ [1–4]. These are conservation and bookkeeping results. They are not hardware specifications validated by test.

### 2.2 The principal physical blockers

The programme is dominated by coupled blockers, not by a shortage of conceptual diagrams.

- **Moving interface.** No experiment demonstrates that the complete moving article can tolerate the acceleration, electromagnetic coupling, release transient and thermal environment across the reference envelope. The nearest prior art—solid-armature railguns—documents contact transition and rail damage near  $2 \text{ km s}^{-1}$  [52], three orders of magnitude below the reference release speed.
- **Pulsed power.** Bulk energy, pulse power, switching, transmission, synchronisation and cell coupling are separate requirements. Satisfying one does not satisfy the others. The engineering envelope is published: the largest operating pulsed-power facility stores tens of megajoules [53], capacitor specific energy and life are bounded by a mature literature [54], and the highest demonstrated direct-current field is 45.5 T in a miniature insert [55].
- **Structure and reaction.** Momentum must leave the launcher–platform system. A rear valve or counter-mass can export opposing momentum, but it does not erase transient loads. For the lower-speed T-DART case already analysed, 3 MN s on a 2,000 t platform gives  $1.5 \text{ m s}^{-1}$  uncompensated motion; ideal matching with  $4.5 \text{ km s}^{-1}$  expelled mass requires 666.7 kg per event [3,4].
- **Flight survival.** The flight path is not empty. Dust, residual gas, impact plasma and charge exchange are trajectory-dependent hazards. The published fracture calculation is intentionally a lower-bound crack-energy screen, not a ballistic limit. A 50 mm cross-section also sweeps about 20.3 kg of atmospheric column in a vertical Earth passage; an intact atmospheric mode is therefore excluded [1,3,4].
- **Materials.** Boron-doped diamond remains a functional-material candidate, not a selected monolithic structure. CVD/PCD diamond, diamond–SiC, toughened SiC, boron carbide, tungsten, carbon-rich sacrificial layers and graded hybrids require matched dynamic tests [1,4].
- **Assembly and control.** A seven-kilometre flexible structure is far beyond ordinary spacecraft integration. Environmental magnetic torque can assist slow momentum management in LEO but does not provide the required rapid reference slew [6].

These blockers determine the build order.

## 3. Technology and market reality

### 3.1 Batteries are the buffer tier

Tesla currently lists a 4-hour Megapack at 3,916 kWh and 979 kW, or approximately 14.10 GJ of stored energy [9]. A simple energy-equivalent count is

$$N_{MP} = \frac{514.286 \text{ GJ}}{14.098 \text{ GJ}} = 36.48. \quad (3)$$

Using 36.48 equivalent units for comparison, not procurement, gives 35.7 MW combined output. The mean admitted-power screen is

$$\dot{P}_{admit} = \frac{428.571 \text{ GJ}}{0.014 \text{ s}} = 30.612 \text{ TW}, \frac{\dot{P}_{admit}}{P_{MP, fleet}} \approx 8.57 \times 10^5. \quad (4)$$

The conclusion is not that batteries are useless. Commercial grid batteries can recharge, buffer and stabilise a slow primary-energy tier. They cannot directly serve as the millisecond pulse tier. The power architecture must be hierarchical: grid or generation; electrochemical buffer; intermediate pulsed store; switching and pulse compression; local module; launcher cell. Each boundary requires a measured efficiency, thermal ledger, fault-containment case and replacement strategy.

### 3.2 Public pulsed-power comparators

Sandia reports that Z stores up to 22 MJ and can deliver approximately 80 TW to an experiment on a roughly 100 ns scale [7]. Its Marx banks, intermediate water capacitors, pulse-forming lines and triggered switches illustrate why stored energy and delivered pulse are different engineering layers [8]. Sandia has also discussed a possible 130 MJ stored-energy facility for future magnetically driven high-energy-density research [30].

The ARGUS reference nameplate is

$$R_z = \frac{514.286 \text{ GJ}}{22 \text{ MJ}} = 2.3377 \times 10^4 \quad (5)$$

times Z by stored energy. The comparison is not like-for-like in pulse shape, load or topology. It nevertheless eliminates the idea that an existing capacitor catalogue can be multiplied directly into a complete solution. The right question is whether a progressively scaled cell-and-string architecture can preserve efficiency, timing, insulation, force transfer and fault containment as energy grows.

### 3.3 Launch and in-space assembly

SpaceX describes Starship as designed to carry more than 100 metric tonnes to orbit in a fully reusable configuration [15]. That capacity materially changes the logistics of large spacecraft but does not close ARGUS mass. Neither the active launcher nor its energy, thermal, shielding, structure, robotics and consumables have a credible integrated mass estimate.

NASA lists the International Space Station at 419,725 kg with a 94 m truss [16]. That comparison should not be used as a linear cost or mass model, but it exposes the unprecedented length and integration class of a seven-kilometre article. NASA's 2025 in-space servicing, assembly and manufacturing survey documents a growing ecosystem rather than a mature kilometre-class supply chain [17]. SpiderFab and related concepts show that on-orbit fabrication can improve stiffness-to-mass and stowed volume, but they are technology

directions, not qualification evidence for a pulsed-power structure [18]. NASA's cancellation of OSAM-1 is an equally important negative precedent: ambitious assembly programmes can fail on cost, schedule and integration despite strong strategic logic [19].

The build strategy must therefore separate:

1. a non-energetic structural and robotic assembly pathfinder;
2. a ground-based energetic demonstrator;
3. a later orbital energy and thermal demonstration; and
4. any mission-authorised civil operation.

## 4. Outcome architecture

### 4.1 What “success at year eight” means

The year-eight outcome is a verified evidence package and integrated demonstrator, not a hidden redefinition of “operational”.

| Metric                | Year-eight success threshold                                                                                     | Why it matters                                                               |
|-----------------------|------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| critical technologies | independent assessment at approximately TRL 5–6 for the selected subscale architecture                           | NASA typically expects TRL 6 or higher before flight-system integration [14] |
| moving interface      | repeatable test series with measured force, heating, release and damage distributions                            | replaces ideal force-density inference                                       |
| pulse system          | representative module and physical string with measured energy, power, timing, efficiency and fault isolation    | tests the 6,000-bay architecture                                             |
| integrated facility   | reduced-energy ground demonstrator with closed reaction, thermal and safety ledgers                              | proves system interaction                                                    |
| materials             | down-selected layered projectile architecture validated over reachable test speeds                               | addresses dust, flaws and interfaces                                         |
| orbital pathfinder    | non-energetic modular structure with assembly, metrology, repair and safe-state control                          | isolates ISAM risk                                                           |
| governance            | enacted authority, licensing/supervision plan, independent authorisation and international consultation protocol | makes civil intent verifiable                                                |
| cost                  | refreshed WBS and joint cost-and-schedule confidence for the next phase                                          | prevents commitment by narrative                                             |

NASA defines TRL 6 as a high-fidelity model or prototype demonstrated in a relevant environment with critical scaling issues addressed [14]. A subscale ARGUS system can reach that level only for the architecture actually tested. The rating must not be transferred automatically to the full seven-kilometre system.

## 4.2 What “first deployable capability” means

If Gate G4 closes, years 9–16 can pursue full-scale qualification and a first supervised civil capability. “Deployable” requires:

- a mission class and safe operating envelope;
- independently verified physical and software configuration;
- range, trajectory and casualty-risk analysis held in controlled mission records rather than an open paper;
- launch, registration, licensing and continuing supervision;
- dual authorisation for any irreversible energetic action;
- safe state, abort, inspection and post-event accountability;
- international threat validation and consultation for planetary defence;
- a funded decommissioning and disposal plan.

No single date can be credible before the G4 mass and energy baseline. The central schedule is 12–16 years for a first civil capability. Eight years remains an aggressive earliest deployment decision. Full reference-scale performance may take longer or may be rejected by evidence.

## 5. Build process and industrial organisation

### 5.1 Workstream decomposition

The build should be organised around nine accountable workstreams.

| Workstream              | Build responsibility                                           | Primary gate output                                                  |
|-------------------------|----------------------------------------------------------------|----------------------------------------------------------------------|
| programme and assurance | sponsor authority plus independent review board                | controlled requirements, hazards, configuration and decision records |
| digital engineering     | systems integrator with national laboratories and universities | traceable model, test data, uncertainty and verification ledger      |
| projectile materials    | materials laboratories and specialised suppliers               | ranked layered architectures and dynamic failure statistics          |
| pulsed power            | national laboratory–prime contractor team                      | representative store, switch, transmission and cell chain            |



| Workstream                | Build responsibility                                  | Primary gate output                                                         |
|---------------------------|-------------------------------------------------------|-----------------------------------------------------------------------------|
| structure and reaction    | civil/space prime plus structural test centre         | transient-load, thermal, vibration and momentum-export closure              |
| ground demonstrators      | dedicated test-site authority                         | progressively integrated strings and safe high-energy testing               |
| ISAM pathfinder           | spacecraft/robotics prime and launch provider         | modular assembly, metrology, inspection and repair                          |
| flight-science pathfinder | civil space agency and international science partners | non-operational materials/environment experiment                            |
| operations and governance | public mission authority                              | secure control, audit, supervision, consultation and liability arrangements |

## 5.2 Make, buy and partner

The programme should buy commercial maturity where it exists and retain public control where consequences are irreversible.

- **Commercially buy:** conventional generation; Tesla-class or equivalent buffer storage; standard launch services; data-centre hardware; conventional spacecraft buses; cloud-isolated development environments; non-sensitive robotics and metrology.
- **Co-develop:** intermediate pulse storage; switching; superconducting or alternative cell technology; high-rate diagnostics; dynamic material architectures; autonomous large-structure assembly; high-reliability thermal and power distribution.
- **Publicly own or control:** safety case; energetic test facility; independent verification; mission authorisation; controlled trajectory analysis; audit records; civil command authority; legal supervision; incident investigation.

This division gives companies a commercial market for enabling technologies without outsourcing sovereign responsibility.

## 5.3 Facility sequence

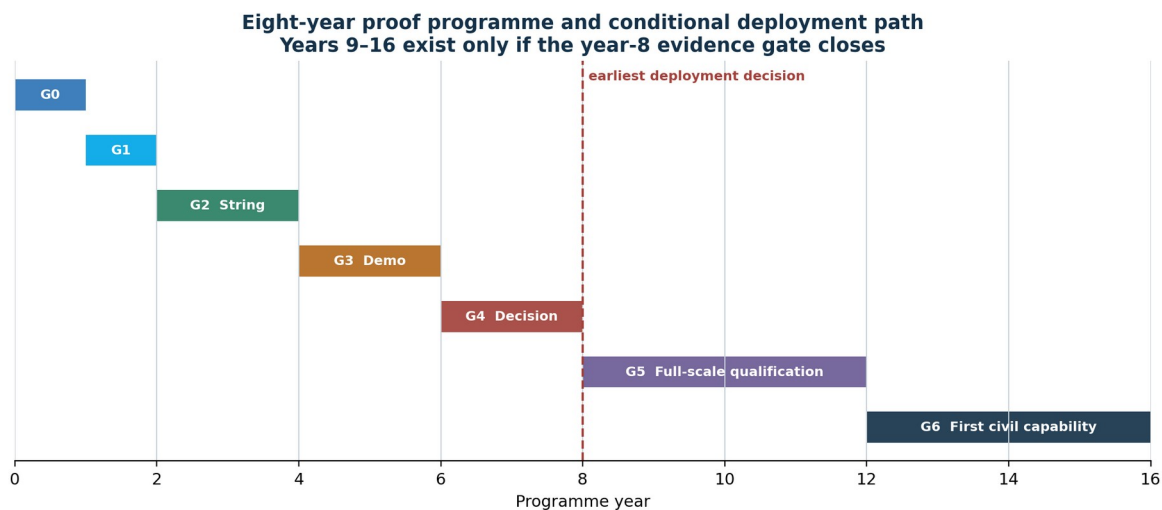
The physical sequence should move from cheap falsification to expensive integration.

1. **Coupons and digital twins.** Calibrate electromagnetic, thermal, fracture, impact and flexible-body models against small experiments.
2. **Representative cell.** Demonstrate measured force and energy transfer into a moving interface with failure containment.

3. **Physical string.** Establish timing, switching, interference, structural reaction and diagnostic observability across multiple cells.
4. **Integrated ground article.** Couple energy buffer, pulse tier, string, vacuum, structure, reaction management, thermal recovery, release and independent safety controls.
5. **Non-energetic orbital structure.** Validate packaging, launch, autonomous assembly, metrology, flexible-body control, repair and disposal without an energetic payload.
6. **Relevant-environment demonstrator.** Scale only after earlier evidence closes; retain an independent authority capable of stopping the programme.
7. **Mission-specific civil deployment.** Proceed only after G4/G5 technical and governance approvals.

The sequence protects the sponsor from a common megaproject failure mode: building an impressive facility before discovering that its core interface cannot meet the required physics.

## 6. Stage-gated schedule



*Eight-year proof programme and conditional deployment path. The first five gates run from formulation to a year-eight deployment decision; years nine to sixteen are conditional on the earlier evidence.*

### 6.1 Gates G0–G4: years 0–8

| Gate | Window | Decision evidence                                                                | Stop or redirect condition                                    |
|------|--------|----------------------------------------------------------------------------------|---------------------------------------------------------------|
| G0   | 0–1    | credible architecture, authority, hazards, WBS, test plan and independent review | no coherent legal authority, safety boundary or cost basis    |
| G1   | 1–2    | calibrated component models; ranked material, pulse and interface candidates     | all candidate branches miss threshold by a predeclared margin |

| Gate | Window | Decision evidence                                                              | Stop or redirect condition                                                         |
|------|--------|--------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| G2   | 2–4    | repeatable representative cell/string data and fault containment               | efficiency, damage, timing or force transfer does not scale                        |
| G3   | 4–6    | integrated subscale closure of energy, heat, structure, reaction and release   | integrated result contradicts component predictions                                |
| G4   | 6–8    | relevant-environment ground demonstrator plus non-energetic orbital pathfinder | mass, survival, cost, governance or assembly cannot support deployment formulation |

NASA's life-cycle model separates phases with Key Decision Points at which a decision authority determines readiness to proceed [11,12]. The ARGUS gates adopt that discipline but are not represented as NASA approval.

## 6.2 Gates G5–G6: conditional years 8–16

G5 authorises full-scale qualification only after a credible mass and energy baseline exists. G6 authorises a first civil capability only after mission-specific safety, legal and international review. The distinction is essential. The sponsor may reach year eight with a valuable national laboratory and no rational case for orbital energetic deployment. That is a successful research outcome if the negative result is established before larger sunk costs.

The schedule has real precedent at both ends. DART demonstrated a planetary defence kinetic impact for a projected life-cycle cost of \$330.6 million, but it was a comparatively small spacecraft and mission [21,22]. NEO Surveyor traces from a 2016 concept study through reformulation and a baseline launch no later than June 2028; its current agency baseline commitment is \$1.6 billion [20]. A genuinely unprecedented integrated system should not be scheduled more confidently than those missions.

The mitigation literature also constrains what the eight-year proof may claim to be *for*. The decadal assessment [44] and the slow-push alternatives [45,46] establish that, for long-warning and well-characterised targets, gravity tractors and ion-beam shepherds are preferable on controllability grounds, and the post-DART mission studies [47,48] define the demonstrated kinetic baseline this programme would have to beat. The engineering prior art points the same way: launch-to-space railgun studies [49,50] were never demonstrated, and the Navy programme was curtailed without fielding [51]. The programme case in this paper survives those facts only as a proof-and-decision programme—establishing whether the extreme-velocity regime is reachable at all—never as a substitute for the demonstrated portfolio. G5 and G6 encode that restriction.

## 7. Cost exposure

### 7.1 Method and limitations

GAO states that reliable estimates should be comprehensive, well documented, accurate and credible [13]. NASA cost practice combines estimating methods with risk and uncertainty; its Joint Cost and Schedule Confidence Level integrates cost, schedule, risk and uncertainty rather than quoting one deterministic number [10,31].

The academic megaproject literature is blunter than either agency handbook, and I adopt its discipline rather than resist it. Flyvbjerg's survey of megaproject performance reports systematic cost underestimation and benefit overestimation across sectors, driven by optimism bias and strategic misrepresentation rather than by bad arithmetic [41]; Merrow's study of industrial megaprojects reports similar base rates for first-of-a-kind process facilities [42]. A reader applying reference-class thinking would therefore treat any single-point estimate for an unprecedented orbital pulsed-power installation as close to meaningless. That is the correct prior, and it is why this section publishes a risk-loaded distribution with explicit triangular inputs and a correlated programme factor rather than one number, why the P50 and P80 are reported rather than the mode, and why every input is labelled a planning assumption open to replacement by quotation—the treatment of uncertainty the decision-analysis canon prescribes [56] and the sensitivity-practice literature audits [57]. The workstream decomposition itself follows the design-methodology mainstream: candidates are retained as sets and narrowed at gates rather than converged early [60], the programme trade space is explored before commitment [58], and the coupled cost-mass-energy propagation is the standard multidisciplinary structure [59]. Where the reference-class critique and this model still disagree—the critique implies even the P80 may be optimistic for a programme with no predecessor class at all—the stage-gate structure of Section 6, in the technology-readiness sense of Mankins [43], is the answer: no tranche commits funds beyond the next measured gate.

This paper supplies an early screening model:

$$C = F_c \sum_{i=1}^9 \text{Triangular}(C_{i,L}, C_{i,M}, C_{i,H}), F_c \sim \text{Triangular}(1.00, 1.15, 1.50). \quad (6)$$

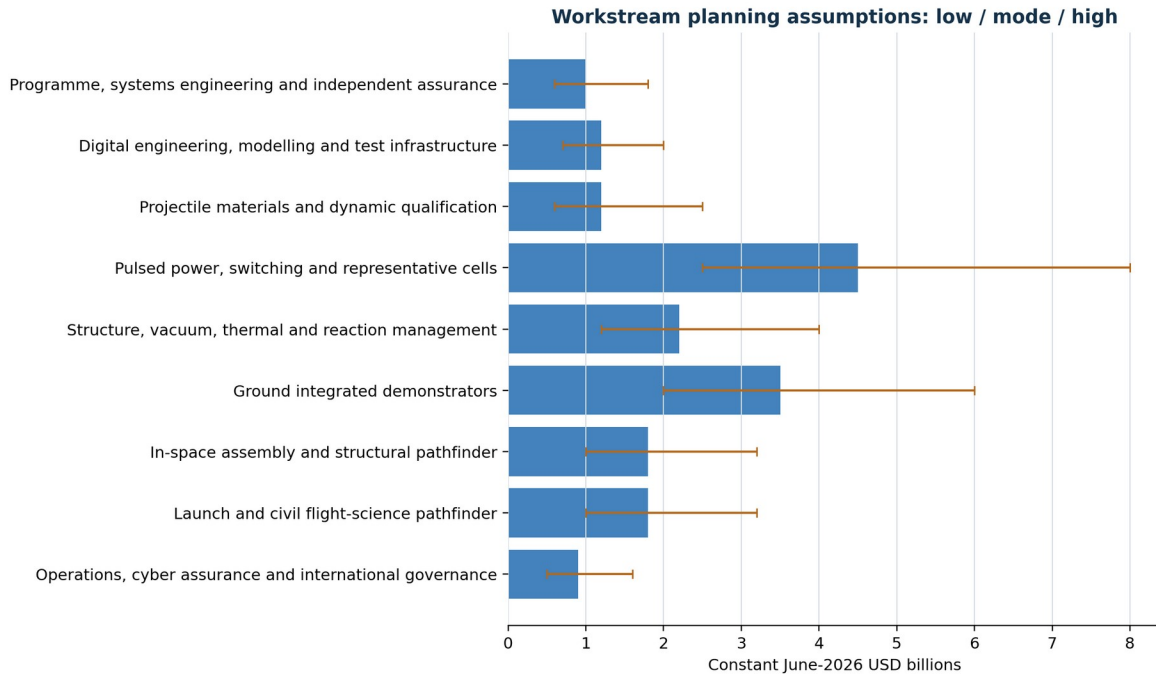
The nine work packages are defined to be mutually exclusive direct costs. The common factor  $F_c$  then represents programme-wide integration, reserve, schedule and supplier-learning uncertainty. Its minimum is unity: it cannot reduce the direct-cost draw. Constant June-2026 dollars exclude future escalation, so inflation is not loaded again into  $F_c$ . The factor prevents the false confidence that would result from treating all workstreams as independent, but it is not a calibrated joint cost-and-schedule model. The reported percentile is

$$C_{pq} = \inf \{c : \Pr(C \leq c) \geq q\}. \quad (7)$$

The inputs are my own planning assumptions informed by public comparator programmes, not supplier bids. Tesla's current order page publishes power, energy and physical specifications but displays price only after project quantity, site and delivery inputs; it also excludes taxes and installation [9]. High-energy capacitors, switches and representative launcher cells likewise require a defined duty cycle and supplier requests for quotation. The estimate therefore does not multiply a guessed retail battery or capacitor price. It excludes a full operational deployment, financing costs, future escalation beyond constant June-2026 dollars, classified infrastructure, war-reserve stocks and operations after the proof programme.

## 7.2 Workstream assumptions

| Workstream                                   | Low | Mode | High | Scope note                                  |
|----------------------------------------------|-----|------|------|---------------------------------------------|
| programme, systems engineering and assurance | 0.6 | 1.0  | 1.8  | independent review and control              |
| digital engineering and test infrastructure  | 0.7 | 1.2  | 2.0  | models, diagnostics, data and facilities    |
| projectile materials and qualification       | 0.6 | 1.2  | 2.5  | multi-material dynamic campaign             |
| pulse power, switching and cells             | 2.5 | 4.5  | 8.0  | dominant facility and scale-up risk         |
| structure, vacuum, thermal and reaction      | 1.2 | 2.2  | 4.0  | distinct transient-load infrastructure      |
| integrated ground demonstrators              | 2.0 | 3.5  | 6.0  | integration excluding cell-development cost |
| in-space assembly pathfinder                 | 1.0 | 1.8  | 3.2  | non-energetic orbital assembly              |
| launch and flight-science pathfinder         | 1.0 | 1.8  | 3.2  | spacecraft, launch and mission assurance    |
| operations, cyber and governance             | 0.5 | 0.9  | 1.6  | secure control and supervision              |



*Illustrative programme cost by workstream. Bars show mode assumptions and whiskers show the low-to-high planning range.*

The revised workstream boundaries remove two forms of double loading in the previous draft: pulse-cell development is no longer repeated inside the ground-demonstrator line, and orbital assembly is separated from the civil flight-science mission. Workstream high cases remain visible, but a second unbounded contingency is not hidden inside each mode.

Public analogues bound scale; none is a parametric cost-estimating relationship for ARGUS:

| Public analogue                         | Published cost  | Approx. June-2026 USD | Relevance and limit                                                                 |
|-----------------------------------------|-----------------|-----------------------|-------------------------------------------------------------------------------------|
| DART projected lifecycle                | \$0.331bn       | not converted         | lower flight-demonstration comparator; mixed-year lifecycle dollars [21]            |
| NEO Surveyor baseline                   | \$1.60bn (2022) | \$1.83bn              | planetary-defence flight project, not an energetic launcher [20,40]                 |
| OSAM-1 rebaseline                       | \$2.05bn (2022) | \$2.34bn              | in-space servicing/assembly scale and overrun warning [38,40]                       |
| National Ignition Facility construction | \$3.50bn (2009) | \$5.45bn              | unique high-energy research facility with a different mechanism and mission [39,40] |

The CPI normalisations use the BLS CPI-U annual averages of 214.537 for 2009 and 292.655 for 2022, and the June 2026 index of 333.952 [40]. They are transparent scale comparisons, not aerospace escalation indices.

7.3 Reconciliation with the earlier \$18.1 billion estimate

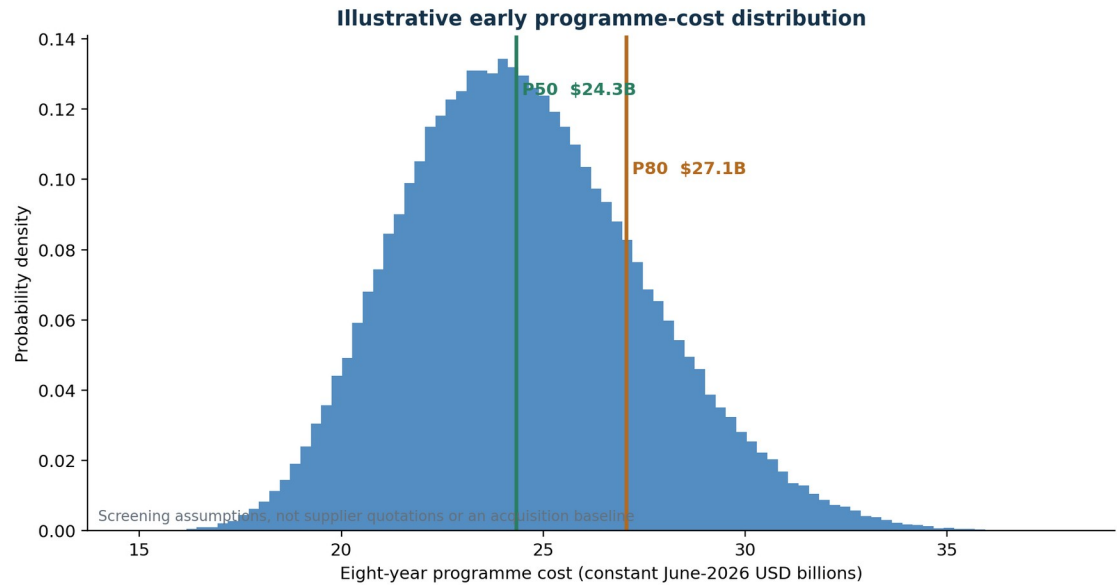
The source archive contains an earlier \$18.1 billion figure. It was correctly rejected as a lifecycle cost for that paper’s obsolete 49 km architecture because it omitted material scope. This audit did not reinstate that old cost object. Starting again from the controlled fixed seven-kilometre G0–G4 proof scope, the revised mutually exclusive work packages independently sum to the same \$18.1 billion at their modes. The numerical convergence is useful, but it is not validation of the archived estimate.

The distinction is:

- **\$18.1 billion:** direct most-likely work-package sum in constant June-2026 dollars;
- **\$24.3 billion P50:** median risk-loaded eight-year proof-programme screen;
- **\$27.1 billion P80:** early confidence target for tranche planning; and
- **unpriced:** full operational seven-kilometre deployment, future escalation, financing, classified infrastructure and post-proof operations.

The raw workstream sums are \$10.1 billion low, \$18.1 billion mode and \$32.3 billion high. With the common factor, the explicit risk-loaded mathematical input envelope is \$10.1–48.45 billion. A deterministic 250,000-sample run with the published seed gives:

| Statistic | Eight-year proof programme |
|-----------|----------------------------|
| P10       | \$20.8 billion             |
| P50       | \$24.3 billion             |
| mean      | \$24.5 billion             |
| P80       | \$27.1 billion             |
| P90       | \$28.6 billion             |



Illustrative early cost distribution for the eight-year proof programme, with P50 and P80 markers.

The relatively narrower central distribution does not invalidate the much wider mathematical envelope. It reflects the triangular sampling assumptions, which themselves require replacement. A decision authority should manage the programme to a confidence target such as P80 only after a real schedule logic network and correlated risk register exist.

7.4 Funding profile and tranche decisions

For an illustrative P80 programme, annual funding is

$$B_y = f_y C_{P80}, \sum_{y=1}^8 f_y = 1. \tag{8}$$

| Programme year | Fraction | P80 annual value |
|----------------|----------|------------------|
| 1              | 3%       | \$0.81 billion   |
| 2              | 7%       | \$1.89 billion   |
| 3              | 12%      | \$3.25 billion   |
| 4              | 16%      | \$4.33 billion   |
| 5              | 18%      | \$4.87 billion   |
| 6              | 18%      | \$4.87 billion   |
| 7              | 15%      | \$4.06 billion   |
| 8              | 11%      | \$2.98 billion   |



Those values should not be appropriated as one unconditional block. The programme should use tranche ceilings:

- **Formulation tranche:** 18 months, ceiling of approximately \$1.0–1.8 billion.
- **Laboratory tranche:** to G2, released only after G1 candidate selection.
- **Integration tranche:** to G3, released only after representative data.
- **Relevant-environment tranche:** to G4, released only after integrated closure.

Years 9–16 require a new cost estimate at G4. The system mass and launch architecture are too immature for an honest deployment point estimate. As a stress test, a sponsor should assume exposure comparable to a major national space campaign: NASA's contributions to Artemis were projected at \$93 billion for fiscal years 2012–2025 [29]. That comparison defines scale, not cost.

## 8. Control potential after deployment

### 8.1 Control is broader than command

The relevant question is not whether a control room can transmit a command. It is whether public authority can constrain the complete lifecycle: requirements, procurement, configuration, launch, operation, irreversible action, inspection, incident response and disposal.



*Governance stack for a high-consequence civil space capability, from public authority to continuing supervision.*

| Layer            | Required control                              | Evidence                            |
|------------------|-----------------------------------------------|-------------------------------------|
| public authority | explicit statutory mandate, budget authority, | published authority and accountable |

| Layer                     | Required control                                                                                 | Evidence                                                                           |
|---------------------------|--------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
|                           | licensing jurisdiction and treaty responsibility                                                 | minister/agency                                                                    |
| independent authorisation | technical, legal and mission approval separated from routine operations                          | recorded decision with dissent and conditions                                      |
| dual authorisation        | two-person approval for critical or irreversible operations                                      | NIST recognises dual authorisation as risk reduction for sensitive operations [23] |
| configuration assurance   | signed software/hardware baseline, supply-chain provenance and controlled change                 | reproducible build, attestation and independent verification                       |
| safe state                | de-energised, passively stable or otherwise bounded condition after fault or lost communications | demonstrated fault response                                                        |
| telemetry and audit       | independent observation, immutable logs and post-event reconstruction                            | external telemetry and retained evidence                                           |
| continuing supervision    | inspection, licence conditions, incident reporting, sanctions and decommissioning                | Article VI-compatible national supervision [5,25]                                  |

The architecture should keep detailed command paths, credentials, operational trajectories and target solutions outside the open publication record. Transparency should concern authority, constraints, evidence and accountability—not the disclosure of exploitable operational data.

## 8.2 International control for planetary defence

Planetary defence is inherently a shared-risk activity. NASA's strategy calls for transparency, multilateral cooperation and international engagement [20,26]. The International Asteroid Warning Network supports detection and characterisation; the Space Mission Planning Advisory Group provides a forum for space agencies to collaborate on mitigation options [27]. An ARGUS mission should therefore require:

1. independent confirmation of the threat;
2. shared target-state and uncertainty review;
3. consultation on the proposed intervention and alternatives;
4. allocation of registration, responsibility and liability;
5. notification, observation and post-event data release; and
6. pre-agreed abort and dispute procedures.

The legal companion paper explains why Outer Space Treaty Article VI imposes state responsibility and supervision but does not itself create a universal licensing or sanctions power [5]. Domestic legislation must supply the actual authority.

## 9. The strategic-advantage hypothesis

### 9.1 The race already exists

Space control is not a speculative motive invented for this project. It is a declared national and alliance objective expressed in different political vocabularies.

| Actor          | Published objective                                                                                           | Competitive implication                            |
|----------------|---------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| United States  | space superiority and freedom to operate without prohibitive interference [24,28]                             | explicit control objective                         |
| China          | a “new journey towards a strong space presence” and comprehensive space infrastructure [33]                   | sustained state-led industrial competition         |
| European Union | resilience, protection, threat response, dual-use capability and a stronger role as a global space power [34] | strategic autonomy and collective control          |
| United Kingdom | become a meaningful space actor and preserve strategic advantage with allies [35]                             | civil-defence integration and sovereign capability |
| NATO           | space as an operational domain essential to deterrence and defence [36]                                       | allied demand for assured services and resilience  |
| India          | augment national capability and create a thriving commercial space ecosystem [37]                             | widening industrial and launch competition         |

These sources do not show that every actor seeks identical military “supremacy”. They show convergence on the underlying requirements: assured access, protected services, industrial depth, autonomous decision-making, domain awareness and influence over rules. That convergence is a strategic race. A state that waits until another actor controls the launch, construction, data and governance stack must buy access on terms it did not set.

For an established or aspiring space power, non-participation carries its own costs. It risks dependence on another actor’s launch cadence, orbital infrastructure, sensing, technical standards and crisis decisions. The case for funding ARGUS research therefore rests less on the machine itself than on the credible position it buys while there is still time to shape the industrial and legal architecture of that wider race. A programme that fails its physics gates can be stopped, but competence a state has never built cannot be recovered on demand.

ARGUS is one possible way to accelerate several layers of that stack at once. Its attraction is concentration: one programme forces progress in energy, pulsed power, materials, heavy launch, robotic construction, sensing, high-assurance software and legal control. If the architecture scales modularly, it may produce strategic option value faster and at lower cost than building the same industrial capabilities as unrelated programmes. That is a hypothesis to be tested at G0–G4, not assumed in the financial model.

The programme is not cheap in ordinary terms. Its \$27.1 billion P80 proof screen is, however, below the cited \$93 billion scale of NASA's Artemis contribution through fiscal year 2025 [29], while purchasing a portfolio of capabilities relevant to civil exploration, national security and commercial infrastructure. The investment argument is one of comparative affordability.

## 9.2 A testable definition

United States Space Force doctrine defines space superiority as the degree of control that permits operations at a chosen time and place without prohibitive interference while denying the same to an adversary [24]. A 2025 United States policy statement likewise presents space superiority as linked to national economic and security interests [28]. These sources make the strategic hypothesis legitimate to examine. They do not prove that ARGUS would deliver it.

The hypothesis can be decomposed:

| Necessary condition for broad superiority | ARGUS contribution                                                | Current assessment                    |
|-------------------------------------------|-------------------------------------------------------------------|---------------------------------------|
| persistent awareness                      | sensor, tracking, metrology and data-fusion spillovers            | credible programme benefit            |
| assured access and logistics              | launch demand, modular assembly, servicing and repair             | potentially major, dependent on ISAM  |
| resilient operations                      | distributed modules, repair and alternative modes                 | conceptual; architecture unvalidated  |
| protected command                         | high-assurance control and independent telemetry                  | designable; must be demonstrated      |
| freedom from prohibitive interference     | one large structure is observable and potentially vulnerable      | not established                       |
| ability to deny adversary interference    | outside the civil programme and not established by these papers   | no evidentiary support                |
| alliance and legal legitimacy             | transparent planetary-defence service and shared governance       | achievable if designed from inception |
| industrial depth                          | pulse power, materials, robotics, energy and launch supply chains | strongest near-term strategic pathway |

The strict claim—"ARGUS gives one state military supremacy in space within eight years"—is not supported.

The stronger evidence-based claim is:

*A sponsor that leads the enabling industrial base, assembly logistics, sensing, standards and governed planetary-defence service could gain durable strategic space advantage and significant agenda-setting power.*

### 9.3 Strategic outcomes ladder

| Level | Outcome                                                                           | Earliest defensible horizon         |
|-------|-----------------------------------------------------------------------------------|-------------------------------------|
| 0     | reproducible science and falsification                                            | present publication set             |
| 1     | national laboratories, skills and supply-chain learning                           | years 1–4                           |
| 2     | integrated civil-space demonstrator and independent control regime                | years 5–8                           |
| 3     | coalition-standard leadership in large-space infrastructure and planetary defence | years 6–12                          |
| 4     | first supervised civil capability with strategic option value                     | conditional years 12–16             |
| 5     | broad space superiority                                                           | cannot be attributed to ARGUS alone |

The programme can contribute to a wider space-superiority posture through industrial capacity, resilience and infrastructure control. It cannot replace launch fleets, communications, navigation, intelligence, alliance access, cyber defence, manoeuvre, distributed architectures or political legitimacy. The ladder exists to stop level 5 being claimed on the strength of level 1.

### 9.4 Why overt weapon framing destroys value

An openly counterspace ARGUS proposition would have five predictable effects:

1. it would invite pre-emption against a large, observable and infrastructure-dependent object;
2. it would increase incentives for competitors to build asymmetric countermeasures rather than match the system;
3. it would complicate launch licensing, insurance, financing and international partnership;
4. it would undermine the civil evidence-sharing needed for planetary defence; and
5. it would weaken the rule-of-law claim by concentrating irreversible power without shared verification.

The United Nations General Assembly's 2022 resolution on destructive direct-ascent anti-satellite testing reflects international concern over debris-generating counterspace behaviour [32]. ARGUS is not a direct-ascent missile, but the political lesson transfers: strategic legitimacy depends on observable restraint and responsible behaviour.

The most powerful strategic configuration is therefore not unilateral secrecy. It is sovereign technical leadership embedded in a coalition governance structure: the sponsor owns the difficult industrial capability while rules, threat validation and mission authorisation make misuse politically and technically harder.

## 10. Deployment options

| Option                                        | Eight-year plausibility  | Strategic value                                      | Principal defect                                                                           | Decision                |
|-----------------------------------------------|--------------------------|------------------------------------------------------|--------------------------------------------------------------------------------------------|-------------------------|
| ground national research facility             | high if funded and gated | pulsed power, materials, skills, deterrent knowledge | not an orbital capability                                                                  | baseline                |
| non-energetic LEO assembly/control pathfinder | moderate                 | ISAM, metrology, repair, control and launch cadence  | does not prove energetic system                                                            | baseline                |
| reduced-energy civil flight-science article   | moderate after G3        | environment and materials evidence                   | strict safety and mission limits                                                           | conditional             |
| full seven-kilometre LEO energetic system     | low within eight years   | high option value if it closes                       | mass, power, vulnerability, dust, control and authorisation                                | no commitment before G4 |
| cislunar or heliocentric system               | not schedulable          | less atmosphere/debris exposure for some missions    | logistics, communications, thermal and control complexity; geomagnetic method inapplicable | post-G4 study           |

The geomagnetic companion paper is LEO-specific [6]. It must not be used to support cislunar or heliocentric control. Likewise, the absence of a complete atmosphere beyond low orbit does not mean absence of matter: dust and plasma remain mission variables.

## 11. Scientific and commercial spillovers

The programme has investment value before full system closure.

- **Pulsed power:** switching, insulation, transmission, diagnostics, fault isolation and high-energy-density research.
- **Energy systems:** orchestration of grid storage, intermediate stores, thermal recovery and high-consequence power controls.
- **Materials:** diamond composites, ceramics, refractory layers, high-strain-rate interfaces and impact diagnostics.
- **Space construction:** autonomous assembly, structural metrology, reconfiguration, servicing and inspection.
- **Space-domain awareness:** object characterisation, uncertainty fusion, conjunction analysis and independent telemetry.
- **Planetary science:** impact physics, equations of state, ejecta characterisation and mitigation modelling.

- **Governance technology:** auditable authorisation, configuration attestation and cross-border mission evidence.

Commercial partners should receive intellectual-property and market rights for ordinary civil applications while the sponsor retains safety, mission-authorisation and controlled-technology rights. A public-private special-purpose programme can coordinate those interests, but it should not transfer treaty responsibility to a corporation.

## 12. Risk register and falsification

| Risk                                             | Leading indicator                                                   | Gate response                                         |
|--------------------------------------------------|---------------------------------------------------------------------|-------------------------------------------------------|
| moving article fails before representative speed | damage and release distributions diverge from models                | down-select materials or terminate speed branch       |
| cell/string scaling loses force or efficiency    | measured scaling exponent misses preregistered band                 | re-architect; no length procurement                   |
| pulse tier mass becomes prohibitive              | energy and power specific performance fail independent lower bounds | reduce energy mode or stop                            |
| dust survival cannot close                       | calibrated impact model predicts unacceptable mission loss          | change geometry/material/trajectory or reject mode    |
| structure cannot survive transient loads         | strain, vibration or alignment exceed recoverable bounds            | shorten, segment or terminate architecture            |
| counter-impulse is operationally excessive       | consumables, plume, contamination or timing exceed limit            | alternative momentum management or reject mode        |
| ISAM cannot achieve required precision           | assembly/metrology error exceeds tolerance                          | ground-only facility or redesigned modularity         |
| cost confidence degrades                         | P80 exceeds authorised ceiling                                      | pause and re-scope                                    |
| legal authority or legitimacy fails              | no accountable authorisation and consultation regime                | no orbital energetic deployment                       |
| strategic posture provokes balancing             | partners decline and threat perception rises                        | strengthen civil coalition structure and transparency |

The conditional probability of reaching deployment can be written without inventing numerical confidence:

$$Pr(G_6) = Pr(G_0) \prod_{i=1}^6 Pr(G_i | G_0, \dots, G_{i-1}). \quad (9)$$

The present papers do not supply calibrated values for those conditional probabilities. Publishing a single “chance of success” would therefore be false precision. The purpose of the gates is to learn them.

## 13. Governance and rule of law

This paper stipulates an underlying strategic concern: that increasing occupation and commercial use of space will reshape global power. Infrastructure owners influence access, standards, insurance, data, dispute resolution and the practical meaning of freedom of operation. The rule of law will not be preserved simply because a leading state has the strongest machine. It will be preserved if technical power is coupled to institutions that make decisions reviewable, attributable and constrained.

For ARGUS, that means:

- public law defining the competent authority;
- separation between operator, mission proposer, authoriser and investigator;
- registration and continuing supervision of non-governmental partners;
- independent scientific validation of a planetary-defence threat;
- documented necessity, proportionality and alternatives analysis;
- consultation under the Outer Space Treaty's due-regard framework;
- liability, compensation and insurance arrangements;
- supply-chain, export-control and cybersecurity governance;
- external observation and post-event publication; and
- decommissioning, disposal and environmental responsibility.

The legal companion article should be read alongside this programme paper [5]. Nothing in a DOI deposit constitutes mission authorisation or legal advice.

## 14. Investment case

### 14.1 Why a government should fund formulation

A government is uniquely positioned to value low-probability, high-consequence planetary defence; to build national laboratories; to hold international responsibility; and to sustain a multi-cycle programme. The first tranche is justified if it produces:

- a credible architecture and negative-result capability;
- a national map of relevant suppliers and facilities;
- measured performance of the highest-risk interfaces;
- an independent cost and schedule basis;
- a civil governance model; and
- a portfolio of commercially transferable technologies.



The additional strategic justification is avoiding dependence. Heavy lift, large-structure assembly, high-consequence energy systems and space-domain awareness are likely to shape future access to resources, communications and security. Major powers and alliances are already funding that competition [24,28,33–37]. A formulation tranche buys a sovereign option before a rival or commercial gatekeeper sets the architecture, standards and price of entry.

## 14.2 Why a major launch-and-energy company should participate

A vertically integrated launch, energy, software and manufacturing company would gain a demanding anchor use case for high-cadence heavy lift, large-scale storage, automated fabrication, robotics, power electronics, digital engineering and in-space logistics. Starship-class payload capability is relevant [15], and Tesla-class storage is relevant as a buffer [9], but neither should be presented as proof that the reference architecture closes.

The commercially sensible entry is a scoped consortium role:

1. launch and orbital logistics architecture;
2. non-energetic structural pathfinder;
3. buffer-energy and supervisory-control demonstration;
4. rapid manufacturing and supply-chain design; and
5. options for later phase participation conditional on evidence.

This lets industry create valuable infrastructure without underwriting an unbounded physics claim.

## 14.3 The disciplined optimistic conclusion

The concept is not ready for a production commitment. It is ready for a serious national formulation programme. The difference is not rhetorical. Formulation has bounded cost, named outputs, independent review and stop rules.

A well-run programme could reach a decisive result within eight years. If positive, years 9–16 could produce a first supervised civil capability and a strategically important industrial ecosystem. If negative, the sponsor would have bought high-value science and infrastructure while avoiding a much larger failed deployment.

# 15. Conclusions

The ARGUS-PD/POD-7K research now supports a realistic programme proposition:

1. **Eight years is the proof horizon.** It can plausibly deliver TRL 5–6 evidence for selected subsystems, an integrated reduced-scale ground demonstrator and a non-energetic orbital assembly/control pathfinder.

2. **Twelve to sixteen years is the conditional civil-capability horizon.** It is not guaranteed, and it begins only after the year-eight mass, pulse-power, survivability, assembly, cost and governance gates close.
3. **Commercial batteries help but do not solve the pulse.** The energy count is manageable in tens of Megapacks; the rate gap is approximately  $8.57 \times 10^5$ . A dedicated hierarchical pulse architecture remains the central engineering problem.
4. **The direct mode is \$18.1 billion and the early P80 proof-programme screen is \$27.1 billion.** The former reconciles my own bottom-up arithmetic; the latter retains integration and uncertainty. Neither is suitable for appropriation as an acquisition baseline.
5. **Strategic advantage is credible; unilateral supremacy is not yet evidenced.** The strongest pathway is leadership in industry, launch, ISAM, sensing, logistics, standards and a governed planetary-defence service.
6. **Control and legitimacy are part of the machine.** Independent authorisation, dual control, telemetry, audit, national supervision and international consultation must be designed from the start.

The correct optimistic finding is not that ARGUS will certainly dominate space. It is that a state or coalition willing to fund and govern the programme could acquire a rare combination of scientific knowledge, industrial capacity, strategic option value and institution-building power. That is enough to justify a formulation tranche. It is not enough to justify anything beyond G0, and this paper does not ask for more.

## Data and reproducibility

The accompanying reproducibility package contains `argus_pd_repro/programme_model.py`, `argus_pd_repro/run_programme_analysis.py`, the nine cost-input rows, the 250,000-sample deterministic cost output, annual funding profile, public comparator arithmetic, stage-gate table, figures and unit tests. The model contains no target solutions, firing logic, mission trajectories or live command architecture.

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